BigBasket — Customer Purchase & Market Basket Analysis

A retail customer-behavior and market basket analysis of BigBasket (online grocery) transaction data. The project turns raw transaction records into actionable insights on purchasing behavior, popular categories, product affinities, and customer segments.

Analysis window: March 2011 – December 2014 **Scope:** 62,141 transaction records · 106 customers · 8,387 orders · 1,732 SKUs · 216 product categories

1. Business Objective

Analyze customer transactions and product purchasing patterns to improve retail sales, customer engagement, and cross-selling. Specifically:

- Identify the most frequently purchased product categories
- Measure order volume, purchasing frequency, and repeat-order behavior
- Discover products frequently purchased together (cross-sell / bundling opportunities)
- Analyze purchasing trends over time
- Segment customers by purchasing activity
- Provide data-driven recommendations for marketing and inventory

2. Datasets

File	Role	Records	Notes
`bigBasket.csv`	Primary transaction dataset	62,141	Columns: Member, Order, SKU, Created On, Description
`IMB575-XLS-ENG.csv`	Supporting customer–category dataset	62,141	Member + Description; 56,182 duplicate rows are repeat purchases, not errors
`Memberdata.csv`	Single-customer case study (M09736)	626	62 orders, 127 SKUs — not representative of the full base

Columns (primary dataset): - `Member` — unique customer identifier - `Order` — order identifier - SKU — product (stock-keeping unit) identifier - `Created On` — transaction date/time - `Description` — product category / description

3. Data Cleaning & Preparation

- **Mixed date formats standardized** — `Created On` contained two representations: 36,569 Excel serial numbers (e.g. `41904.94792`) and 25,572 string dates (e.g. `09/04/14 22:56`). All were parsed to proper datetimes with **zero unparseable values**.
- **Validated identifiers** — Member, Order, and SKU counts confirmed (106 / 8,387 / 1,732).
- **Data quality** — zero missing values, zero exact duplicate rows in the primary dataset.
- **Date range** — corrected to March 2011 – December 2014 (41 months of activity).

4. Key KPIs

KPI	Value
Total Orders	8,387
Total Customers	106
Total SKUs	1,732
Total Product Categories	216
Total Product Records	62,141
Items per Order	7.41
Orders per Customer	79.12
Repeat Customer Rate	100%

> ***Not computable:*** *Revenue, average order value (AOV), gross margin, and customer lifetime value (CLV) — the dataset has no price, quantity, or cost fields. This is a* ***descriptive*** *behavioral analysis.*

5. Methodology

1. **Data cleaning** (Python / pandas) — date standardization, validation, duplicate investigation.
2. **KPI computation** — order volume, basket size, frequency, repeat-rate.
3. **Category analysis** — frequency ranking and Pareto (cumulative share).
4. **Market basket analysis** — order-level category baskets; support, confidence, and lift for co-occurring pairs (minimum 2% support).
5. **Customer segmentation** — behavioral RFB model (Recency, Frequency, category Breadth), quartile-scored into four segments.
6. **Time-series trends** — monthly and yearly order patterns.

6. Headline Findings

- **100% repeat customers** — every one of the 106 customers placed multiple orders (avg 79, min 24): a loyal recurring-order base.
- **Fresh produce dominates** — Other Vegetables, Beans, and Root Vegetables are the top 3 categories; the top 10 account for **52.8%** of all records.
- **Strongest affinities (by lift):** Almonds ↔ Cashews (5.63), Snacky Nuts → Namkeen (3.67), and a tightly clustered dal family (Urad / Toor / Moong / Other Dals).
- **Highest-volume pairs:** vegetable staples (Beans, Other Vegetables, Root Vegetables) co-occur in ~24% of orders.
- **Trend:** sharp growth from 1,291 orders (2012) to 3,802 (2013), holding at 3,274 (2014).

Customer Segments (RFB)

Segment	Customers	Profile
Champions	32	Frequent + recent (avg 107 orders, 3.6 days recency) — reward & retain
At-Risk Loyal	21	Frequent historically, ~52 days lapsed — **highest win-back priority**
Promising	21	Recent but lower frequency — nurture to grow
Needs Attention	32	Low frequency + lapsing — re-engage

7. Recommendations

1. **Cross-sell bundles** — nuts (Almonds + Cashews), dal combos (Urad + Toor + Moong), snack pairings (Snacky Nuts + Namkeen).
2. **Win-back campaign** — target the 21 At-Risk Loyal customers.
3. **Inventory focus** — prioritize always-on stock and forecasting for fresh produce (anchors ~24% of baskets).
4. **Grow Promising segment** — frequency incentives and reorder nudges.
5. **Collect price & quantity data** — unlock revenue, AOV, margin, and CLV analysis.

8. Deliverables

File	Description
`BigBasket_Clean_Transactions.csv`	Cleaned transaction dataset (62,141 rows), standardized dates, sorted by customer/order
`BigBasket_Retail_Analytics.xlsx`	8-sheet analytics workbook for Power BI / Tableau: KPIs · Categories · Monthly Trend · Yearly Trend · Customers · Segments · MBA Rules · Top Pairs
`BigBasket_Analysis.pptx`	10-slide insights & recommendations deck
`charts/`	Chart images (top categories, monthly trend, yearly orders, segments, top pairs, Pareto)
`README.md`	This document

9. Constraints & Limitations

- **No pricing/quantity data** — revenue, profit, and margin cannot be measured.
- **Limited coverage** — only 106 customers; results may not generalize to the full business.
- **Retention is a proxy** — repeat-order behavior approximates retention; it is not a cohort-based retention rate.
- **Associations are descriptive** — co-occurrence shows correlation, not causation.
- **Generalization** — historical patterns (2011–2014) may not reflect current preferences.

10. Tools & Reproducibility

- **Python** — pandas, NumPy (cleaning, KPIs, segmentation, market basket)
- **matplotlib** — chart generation
- **xlsxwriter** — Excel workbook
- **pptxgenjs** — PowerPoint deck

Workflow: clean & validate → compute KPIs → category & trend analysis → market basket (support/confidence/lift) → customer segmentation → export workbook + deck.